



DECEMBER 2024

3236244**MANUFACTURING PROCESS - I***Full Marks: 60**Time Allowed: 3 Hours*

Answer to Question No. 1 of Group A must be written in the main answer script.
In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer
Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20

- (i) The point at which the cutting tool reaches, beyond which it will not function satisfactorily until it is reground, is called as: a) Tool wear, b) Tool failure, c) Tool diffusion, d) None of the above.
- (ii) Which cutting condition affects the cutting temperature predominantly?
 a) Depth of cut, b) cutting speed, c) feed, d) All of the above.
- (iii) In a 4-Hi rolling mill, backup rollers are used to _____
 a) Reduce rolling force, b) Increase stiffness of rolling mill, c) Reduce deflection of working rollers, d) Increase reduction in thickness of job.
- (iv) Shaping of metal by application of compressive force in between two or more dies in order to obtain desired shape is done by- a) Forging, b) Rolling, c) drawing, d) blanking.
- (v) In which of the following forging metal is kept in the lower die?
 a) Open die, b) closed die, c) hold die, d) impression die.



- (vi) The metal forming process in blanking operation is associated with which type of force?
 a) Shear force, b) tensile force, c) compressive force, d) all of the above.
- (vii) The energy requirement in direct extrusion is _____ than indirect extrusion.
 a) More, b) less, c) same, d) no correlation.
- (viii) The surface finish of work piece in hot working is _____ the surface finish in cold working.
 a) Better than, b) worse than, c) same as, d) no correlation.
- (ix) By which material is lathe bed made?
 a) High carbon steel, b) cast iron, c) mild steel, d) copper.
- (x) For which kind of chuck is centering not required?
 a) 4-jaw chuck, b) magnetic chuck, c) 3-jaw chuck, d) mandrel.



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- (xi) Which type of feed is needed in facing operation?
a) Longitudinal, b) cross, c) both longitudinal and cross, d) angular.
- (xii) In neutral flame acetylene and oxygen is in _____ by volume.
a) 1:1 ratio, b) 2:1 ratio, c) stoichiometric ratio, d) ratio.
- (xiii) The ability of the moulding sand to withstand the heat of melt without showing any sign of softening is called as _____.
a) Strength of cohesiveness, b) refractoriness, c) collapsibility, d) flowability.
- (xiv) Which of the following is not a requirement of a good pattern material?
a) It should be light in weight, b) it should have smooth surface, c) it should be easily available, d) it should have low strength to break it and remove easily from mould.
- (xv) AC arc in arc welding is used for welding _____.
a) Hard material, b) highly oxidative material, c) ductile material, d) any material.

GROUP-B

2. (i) Classify manufacturing processes. 3
(ii) What is the main advantage of die forging over impact forging? 2
(iii) Discuss about any two Forging Defects. 3
3. (i) Classify rolling mill as per number of rollers. Draw figures for each of them. 3
(ii) Why smaller roller should be used to decrease contact surface for same deformation? 2
(iii) Write differences between direct and indirect extrusion. 3
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4. (i) Briefly discuss about Different components and their functions of standard die set with a punch. 5
(ii) What are punching and blanking operation? 3
5. (i) Draw three views of a single point cutting tool showing different angles, radius and planes in appropriate views. 3
(ii) Show the specifications of a single point cutting tool having nomenclature 8-10-6-6-6-10-0.2. 2
(iii) Name some cutting tool materials. 1
(iv) Mention the factors affecting tool life. 2
6. (i) Specify a centre lathe. 3
(ii) Why Face Plate and Angle Plates are used in Lathe Machine? 2
(iii) Name any three types of lathe operation showing their schematic representation. 3



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7. (i) Draw different types of oxy-acetylene flame showing different zones. 3
(ii) When reverse polarity is used in arc welding? 2
(iii) What are the causes of using flux in arc welding? 2
(iv) Name any one field of application of TIG welding. 1
8. (i) What is match plate pattern? 2
(ii) Name any two types of pattern allowances. 2
(iii) Name the ingredients of moulding sand with their utility. 2
(iv) Write steps in investment casting. 2
9. (i) Discuss any one type of taper turning operation with figure. 3
(ii) Write down field of application of die and centrifugal casting. 2
(iii) Write down basic differences of brazing and soldering. 2
(iv) On what kind of material spot welding can be done? 1
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